

HeatWatch AI

AI-Based Climate Intelligence System for Heatwave Monitoring, Prediction, and Early Warning

Historical Data Report

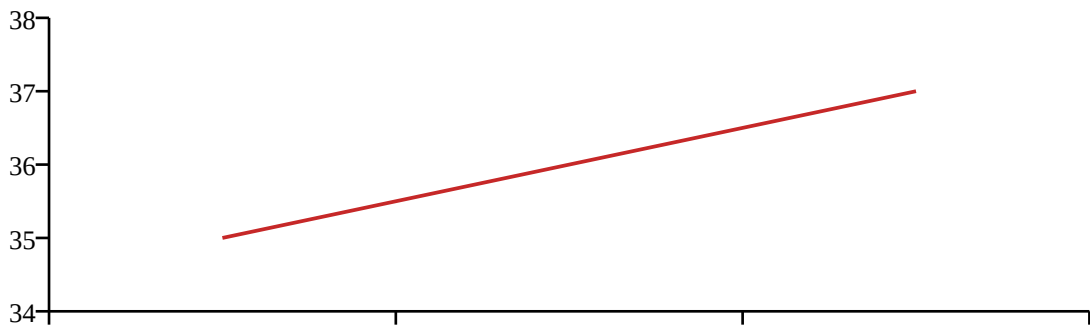
Location	Mumbai, Maharashtra
Coordinates	19.0760, 72.8777
Reporting period	2026-09-12 to 2026-09-19
Generated at (UTC)	2026-09-19 10:32:42
Display timezone	Asia/Kolkata

1. Weather Statistics

Metric	Value	Metric	Value
Observations stored	3	Avg temperature	36.0 °C
Max temperature	37.0 °C	Min temperature	35.0 °C
Avg humidity	60.0 %	Avg wind speed	10.0 km/h
Heat index (avg)	50.0 °C	Heat index (max)	50.0 °C

Heat index is reported only where the NWS calculation is applicable (3 of 3 observations). Data modes present: DEMO.

Temperature trend (stored observations)



2. Heatwave Risk Analysis

Predictions	3
Low risk	0
Moderate risk	0
High risk	3
Average severity	93.4 / 100
Peak severity	93.4 / 100
Data modes	DEMO

Risk assessments combine a weather-based severity model with a documented rule-based sentiment fusion layer. Model methodology and limitations are listed in the HeatWatch AI model card ([docs/model/model_card.md](#)).

3. Sentiment Analysis Summary

Posts analysed	19
Positive / Neutral / Negative	12 / 3 / 4

Heat-related posts	5
Heat-related distress signals	2
Average sentiment score	0.18
Data modes	DEMO

General sentiment is distinct from heat-related distress; the distress count only includes heat-relevant posts with health/impact vocabulary.

4. Alert Summary

Generated at (UTC)	Level	Severity	Status
2026-09-19 10:32	HIGH	93/100	active

Total alerts in period: 1 (high: 1, moderate: 0). All alerts are application-generated warnings, not official advisories.

5. Data Source and Methodology Notes

- Weather observations: provider(s) DEMO; units: temperature in degC, wind in km/h, pressure in hPa.
- Heat index: NWS Rothfusz regression; reported only where applicability conditions hold.
- Sentiment model: VADER (pretrained, English social-media lexicon), compound score in [-1, 1].
- Severity model: weather-based scikit-learn RandomForest trained on the documented heat-index banding dataset, combined with a rule-based sentiment fusion layer. See the model card for evaluation metrics and limitations.
- Provenance: records marked DEMO are seeded demonstration data and do not represent real observations or real public posts.
- Report generated by HeatWatch AI; content is derived exclusively from the database records matching the selected filters.